

# Recall — ratio

What a ratio is, how it differs from a fraction, and how it simplifies.

LESSON 41

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 12 ENTRY

LESSON CLOCK

10'

12'

12'

6'

|                        |        |
|------------------------|--------|
| What a ratio says      | 10 min |
| Simplifying            | 12 min |
| Ratio against fraction | 12 min |
| Reading a ratio back   | 6 min  |

LEARNING OBJECTIVES

- Write a ratio from a description and simplify it.
- Convert units before comparing.
- Say what each number of a ratio counts.
- Distinguish a ratio from a fraction of the whole.

TEXTBOOK

|           |                      |
|-----------|----------------------|
| National  | pp. 109–112          |
| Cambridge | Stage 7, pp. 118–122 |
| Workbook  | Exercise 12.1        |

01

## Explanation

### What a ratio says

A **ratio** compares two quantities of the same kind. In a class of 12 boys and 18 girls the ratio of boys to girls is 12 : 18, which simplifies to 2 : 3.

| SITUATION               | RATIO      | SIMPLIFIED |
|-------------------------|------------|------------|
| 12 boys, 18 girls       | 12 : 18    | 2 : 3      |
| 20 cm to 50 cm          | 20 : 50    | 2 : 5      |
| 4 red, 6 blue, 10 green | 4 : 6 : 10 | 2 : 3 : 5  |

THE ORDER MATTERS

2 : 3 is boys to girls; 3 : 2 is girls to boys. Writing the ratio the wrong way round reverses every answer that follows.

### Simplifying

Divide every part by their highest common factor — exactly as with a fraction. And convert to the same units first.

| RATIO                       | FIRST STEP    | SIMPLIFIED |
|-----------------------------|---------------|------------|
| 12 : 18                     | divide by 6   | 2 : 3      |
| 50 cm : 2 m                 | 50 : 200      | 1 : 4      |
| 30 min : 2 h                | 30 : 120      | 1 : 4      |
| 0.5 : 2                     | 5 : 20        | 1 : 4      |
| $\frac{1}{2} : \frac{1}{3}$ | multiply by 6 | 3 : 2      |

DIFFERENT UNITS MAKE A RATIO MEANINGLESS

50 cm : 2 m is not 25 : 1 — it is 1 : 4. Converting first is not optional; it changes the answer completely.

Ratio against fraction

A ratio compares two parts with each other; a fraction compares one part with the whole.

| STATEMENT                                   | MEANING                            |
|---------------------------------------------|------------------------------------|
| boys to girls = 2 : 3                       | for every 2 boys there are 3 girls |
| boys are $\frac{2}{5}$ of the class         | 2 of every 5 pupils                |
| girls are $\frac{3}{5}$ of the class        | the parts add to 5                 |
| boys are $\frac{2}{3}$ as many as the girls | the ratio again, as a fraction     |

ADD THE PARTS TO GET THE WHOLE

A ratio of 2 : 3 has 5 parts altogether. Every fraction-of-the-whole question starts by adding the parts.

Reading a ratio back

| RATIO     | TOTAL PARTS | IF THE TOTAL IS 30 |
|-----------|-------------|--------------------|
| 2 : 3     | 5           | 12 and 18          |
| 1 : 4     | 5           | 6 and 24           |
| 1 : 2 : 3 | 6           | 5, 10 and 15       |
| 3 : 7     | 10          | 9 and 21           |

ONE PART FIRST, THEN MULTIPLY OUT

Total ÷ number of parts gives one part; each share is that times its own number. Sharing in a ratio, four lessons from now, is exactly this table read forwards.

02 Worked examples

EXAMPLE 1 Simplify the ratio 50 cm : 2 m.

- 1 Convert to the same unit: 50 cm : 200 cm.
- 2 Divide both by 50.
- 3 = 1 : 4

ANSWER

1 : 4

**EXAMPLE 2** A class has 12 boys and 18 girls. Write the ratio of boys to girls and the fraction of the class that is boys.

- ① Ratio:  $12 : 18 = 2 : 3$ .
- ② Total parts:  $2 + 3 = 5$ .
- ③ Boys are  $\frac{2}{5}$  of the class.

ANSWER

$2 : 3$  and  $\frac{2}{5}$

**EXAMPLE 3** Simplify  $\frac{1}{2} : \frac{1}{3}$ .

- ① Multiply both parts by 6.  
A ratio is unchanged by that.
- ②  $3 : 2$

ANSWER

$3 : 2$

### 03 Interactive model

Count the boys and girls in the room and write the ratio on the board; the class simplifies its own numbers, which is more convincing than any textbook example.

**Ratio, or fraction of the whole?**

*12 : 18* simplifies to:

*2 : 3*

*3 : 2*

*6 : 9*

*1 : 2*

*50 cm : 2 m* simplifies to:

*25 : 1*

*1 : 4*

*50 : 2*

*1 : 40*

With *12* boys and *18* girls, boys are what fraction of the class?

$\frac{2}{3}$

$\frac{2}{5}$

$\frac{3}{5}$

$\frac{1}{2}$

A ratio of 2 : 3 has how many parts?

2

3

5

6

2 : 3 and 3 : 2 are:

the same

different

both wrong

equal to  $\frac{2}{3}$

$\frac{1}{2} : \frac{1}{3}$  simplifies to:

1 : 1

3 : 2

2 : 3

1 : 6

30 shared in  $1 : 4$  gives:

6 and 24

10 and 20

5 and 25

15 and 15

30 min : 2 h simplifies to:

15 : 1

1 : 4

30 : 2

1 : 2

## 04 Terminology

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Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH             | O'ZBEKCHA             | РУССКИЙ             |
|---------------------|-----------------------|---------------------|
| Ratio               | Nisbat                | Отношение           |
| Part                | Ulush                 | Часть               |
| To simplify a ratio | Nisbatni qisqartirish | Сократить отношение |
| Equivalent ratios   | Teng nisbatlar        | Равные отношения    |
| Order of the terms  | Hadlar tartibi        | Порядок членов      |
| Same units          | Bir xil birlik        | Одинаковые единицы  |
| Total parts         | Jami ulushlar         | Всего частей        |
| Comparison          | Taqqoslash            | Сравнение           |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.



**05 Quick check****Check yourself**

A ratio compares:

a part with the whole

two quantities of the same kind

two units

nothing

Before simplifying, the units must be:

different

the same

metric

ignored

$20 : 50$  simplifies to:

$2 : 5$

$5 : 2$

$4 : 10$

$1 : 3$

In  $2 : 3$  the first quantity is:





A ratio of  $1 : 2 : 3$  has:





Boys to girls  $2 : 3$  means girls are:

\frac{2}{5}
"/>

\frac{3}{5}
"/>

\frac{2}{3}
"/>

\frac{1}{2}
"/>

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

## Easy · warm the idea up

7 PROBLEMS

1. Simplify  $12 : 18$ ANSWER  $2 : 3$ 2. Simplify  $20 : 50$ ANSWER  $2 : 5$ 3. Simplify  $4 : 6 : 10$ ANSWER  $2 : 3 : 5$ 4. Simplify  $50 \text{ cm} : 2 \text{ m}$ ANSWER  $1 : 4$ 5. Simplify  $30 \text{ min} : 2 \text{ h}$ ANSWER  $1 : 4$ 6. The parts of  $2 : 3$ 

ANSWER 5

7. The parts of  $1 : 2 : 3$ 

ANSWER 6

## Medium · the standard question

7 PROBLEMS

- 12 boys and 18 girls: the ratio

ANSWER 2 : 3

- And the fraction that is boys

ANSWER  $\frac{2}{5}$

- Simplify 0.5 : 2

ANSWER 1 : 4

- Simplify  $\frac{1}{2} : \frac{1}{3}$

ANSWER 3 : 2

- 30 shared in 1 : 4

ANSWER 6 and 24

- 30 shared in 2 : 3

ANSWER 12 and 18

- 30 shared in 1 : 2 : 3

ANSWER 5, 10, 15

## Hard · stretch and reason

7 PROBLEMS

1. Simplify  $750\text{ g} : 1.5\text{ kg}$

ANSWER  $1 : 2$

2. Simplify  $45\text{ s} : 3\text{ min}$

ANSWER  $1 : 4$

3. If boys to girls is  $3 : 7$ , what fraction are girls?

ANSWER  $\frac{7}{10}$

4. A ratio  $2 : 3$  with the first quantity 14: the second

ANSWER 21

5. Simplify  $\frac{2}{3} : \frac{4}{9}$

ANSWER  $3 : 2$

6. Why must units match before simplifying?

ANSWER Otherwise the two numbers count different things

7. A ratio of  $1 : 1$  means

ANSWER the two quantities are equal

## 07 Homework

### Homework — 5 tasks

Convert to the same unit before you simplify anything.

1. Simplify  $16 : 24$ ,  $35 : 49$  and  $18 : 27 : 45$ .
2. Simplify  $80\text{ cm} : 2\text{ m}$  and  $20\text{ min} : 1\text{ h}$ .
3. A class has 14 boys and 21 girls. Write the ratio and the fraction that is girls.
4. Share 40 in the ratio  $3 : 5$ .

5. Explain the difference between “boys to girls is 2 : 3” and “boys are  $\frac{2}{3}$  of the class”.